

Lecture 8 Notes: Aristotle's *Topics*, Book I

Central Theme

Aristotle's *Topics*, Book I introduces dialectical reasoning: reasoning not from scientific first principles, but from reputable or generally accepted opinions.

Topics, Book I = Aristotle's introduction to dialectical reasoning

The *Topics* teaches us how to argue responsibly from what people already accept, especially when we are examining problems rather than demonstrating scientific truths.

1. From Demonstration to Dialectic

The *Posterior Analytics* studied scientific demonstration.

Posterior Analytics = scientific demonstration

The *Topics* studies dialectical reasoning.

Topics = dialectical reasoning

In demonstration, reasoning begins from true, primary, explanatory principles. In dialectic, reasoning begins from generally accepted opinions.

Demonstration begins from first principles.

Dialectic begins from accepted opinions.

Teaching Point

Aristotle is not abandoning logic. He is applying reason to a different kind of setting: argument, inquiry, debate, training, and philosophical examination.

Dialectic works where inquiry has not yet reached demonstration.

2. The Program of the Treatise

Aristotle says that the aim of the treatise is to find a line of inquiry by which we can reason from generally accepted opinions about every problem set before us.

It also aims to help us, when defending a position, avoid saying anything that obstructs our own argument.

Whiteboard

Find arguments from accepted opinions.

Avoid self-obstruction in debate.

Teaching Point

The *Topics* is a practical logical handbook.

It teaches how to find, organize, and use arguments.

3. What Is Reasoning?

Aristotle defines reasoning generally.

Reasoning is an argument in which, certain things being laid down,

something other than these necessarily comes about through them.

This definition connects the *Topics* back to the *Prior Analytics*. Reasoning is still structured necessity.

Teaching Point

The difference between kinds of reasoning lies not in whether they reason, but in the status of their premises.

The kind of reasoning depends on the kind of premises.

4. Four Kinds of Reasoning

Aristotle distinguishes several kinds of reasoning.

1. Demonstrative reasoning
2. Dialectical reasoning
3. Contentious reasoning
4. Mis-reasoning within a special science

Demonstrative Reasoning

Demonstrative reasoning begins from premises that are true and primary, or from premises known through what is true and primary.

Demonstration $\rightarrow$ true and primary premises

Dialectical Reasoning

Dialectical reasoning begins from generally accepted opinions.

Dialectic $\rightarrow$ generally accepted opinions

Contentious Reasoning

Contentious reasoning begins from opinions that merely seem to be generally accepted, or merely seems to reason.

Contentious reasoning $\rightarrow$ seeming acceptance or seeming inference

Mis-Reasoning in a Special Science

Mis-reasoning may occur within a special science when someone reasons from false assumptions that nevertheless belong to that science.

For example, in geometry one may draw a false figure or make an impossible construction.

Scientific mis-reasoning $\rightarrow$ false assumptions within a science

5. Generally Accepted Opinions

Generally accepted opinions are those accepted by:

1. everyone
2. the majority
3. the philosophers
4. all, most, or the most notable philosophers

These are often called *endoxa*: reputable or generally accepted opinions.

Endoxa = reputable opinions

Teaching Point

Aristotle does not mean that popular opinion is automatically true. He means that dialectic begins from premises that an audience or interlocutor can reasonably be expected to grant.

Dialectic tests opinions by reasoning from opinions.

6. Dialectic and Demonstration Compared

Type	Premises	Aim
Demonstration	true and primary	scientific knowledge
Dialectic	generally accepted	examination and argument
Contentious reasoning	apparently accepted	victory or appearance

Teaching Point

Dialectic is weaker than demonstration in certainty, but broader in use.

Dialectic is not science, but it is indispensable for inquiry.

7. The Three Uses of the *Topics*

Aristotle says the treatise is useful for three purposes.

1. Intellectual training
2. Casual encounters
3. The philosophical sciences

Intellectual Training

Dialectic trains the mind by giving us a method for arguing about proposed subjects.

Training → practice in argument

Casual Encounters

Dialectic helps in ordinary discussions. It allows us to meet people on the ground of their own convictions and shift arguments that are unsound.

Conversation → argue from another's beliefs

Philosophical Sciences

Dialectic helps philosophy by enabling us to raise difficulties on both sides of a question. This makes truth and error easier to detect.

It also helps with first principles, because the principles of each science cannot be discussed from within that same science by means of its own principles.

Philosophy → test difficulties and examine principles

Teaching Point

Dialectic is useful because inquiry often begins before demonstration is possible.

Dialectic prepares the way toward principles.

8. Dialectic as Criticism

Aristotle describes dialectic as a critical process.

It is especially useful in relation to the ultimate bases of the sciences, because the first principles of a science cannot be proved from within that science.

Example

Geometry cannot prove its own first principles geometrically, because those principles are prior to geometrical proof.

Teaching Point

Dialectic provides a path toward the principles of inquiry.

Dialectic is the art of testing what demonstration must assume.

9. The Ideal of Dialectical Skill

Aristotle compares dialectic to rhetoric and medicine.

A rhetorician cannot persuade everyone.

A doctor cannot heal everyone.

A dialectician cannot solve every argument in every situation.

But each has an art if he does not omit any available means.

Teaching Point

Dialectical skill means using available argumentative materials well.

Dialectical skill = doing what one can with the available materials

10. Propositions and Problems

Aristotle distinguishes propositions and problems.

Arguments begin from propositions.

Reasonings are about problems.

Proposition = claim offered for acceptance

Problem = question for inquiry

Example of a Proposition

“Animal is the genus of man, is it not?”

Example of a Problem

“Is animal the genus of man or not?”

Teaching Point

The difference between a proposition and a problem is often the turn of phrase.

Dialectic moves between question-form and proposition-form.

11. The Four Predicables

Every dialectical proposition and problem concerns one of four predicables.

1. Definition
2. Property
3. Genus
4. Accident

These are called predicables because they classify how a predicate belongs to a subject.

Every dialectical problem concerns definition, property, genus, or accident.

Teaching Point

The four predicables give the basic map of dialectical problems.

The rest of the *Topics* develops rules for arguing about these.

12. Definition

A definition is a phrase signifying a thing's essence.

Definition = phrase signifying essence

A definition tells what a thing is.

Example

man = animal that walks on two feet

Important Point

A definition is not usually a single word. It is a phrase or formula.

A definition is always an account, not merely a name.

Teaching Point

In the *Topics*, Aristotle is concerned with how definitions function in argument.

A definition can be tested dialectically.

13. Property

A property is a predicate that belongs to a thing alone and is predicated convertibly of it, but does not indicate its essence.

Property = convertible predicate that does not state essence

Example

It is a property of man to be capable of learning grammar.

man $\rightarrow$ capable of learning grammar

capable of learning grammar $\rightarrow$ man

So:

man $\leftrightarrow$ capable of learning grammar

Teaching Point

A property is exclusive, but not essential.

Property belongs only to the subject, but does not say what the subject is.

14. Genus

A genus is predicated in the category of essence of several things differing in kind.

Genus = what something is, said broadly of many species

Example

If asked what a human being is, it is appropriate to answer:

animal

Animal is the genus of man.

man $\rightarrow$ animal

ox $\rightarrow$ animal

Teaching Point

Genus belongs to essence, but it does not give the whole essence.

A genus tells what kind of thing something is, but not exactly what it is.

15. Accident

An accident is what belongs to a thing but is neither definition, property, nor genus.

Aristotle gives a second and clearer account: an accident is something that may belong or not belong to one and the same thing.

Accident = what may belong or not belong to the same thing

Examples

sitting

white

The same person may be sitting now and not sitting later.

The same thing may be white now and not white later.

Teaching Point

Accident is predication without essential necessity.

An accident may be true of a subject without belonging to what it is.

16. Accident, Temporary Property, and Relative Property

Aristotle notes that an accident can become a temporary or relative property.

Temporary Property

If only one man is sitting, then sitting may function as a temporary property of that man.

sitting = temporary property

Relative Property

Two-footed may be treated as a property of man relative to horse and dog, even if it is not a property absolutely.

two-footed = relative property

Teaching Point

Dialectic must notice whether a predicate belongs absolutely, temporarily, or relatively.

Many arguments fail by confusing absolute, temporary, and relative predication.

17. How the Four Predicables Relate

Predicable	Convertible?	States Essence?
Definition	Yes	Yes
Property	Yes	No
Genus	No	Partially
Accident	No	No

Teaching Point

The predicables classify the logical relation between a subject and predicate.

The question is not only what is said, but how it belongs.

18. Sameness

Aristotle distinguishes several senses of sameness.

1. Numerical sameness
2. Specific sameness
3. Generic sameness

Numerical Sameness

Numerical sameness occurs when there is one thing with more than one name.

doublet = cloak

Specific Sameness

Specific sameness occurs when several things belong to the same species.

one man and another man

Generic Sameness

Generic sameness occurs when several things belong to the same genus.

horse and man $\rightarrow$ animal

Teaching Point

Arguments about definition often depend on arguments about sameness and difference.

To test a definition, ask whether the things identified are really the same.

19. Why the Four Predicables Are Exhaustive

Aristotle argues that the four predicables cover all cases of predication.

Every predicate is either convertible with the subject or not.

If Convertible

If the predicate is convertible with the subject, then it is either:

definition

or:

property

It is a definition if it signifies essence.

It is a property if it does not signify essence.

If Not Convertible

If the predicate is not convertible with the subject, then it is either part of the definition or not.

If it is part of the definition, it is genus or differentia.

If it is not part of the definition, it is accident.

Convertible	Definition or Property
Not Convertible	Genus/Differentia or Accident

Teaching Point

The predicables classify all the basic ways a predicate may belong to a subject.

Every predicate has a dialectical status.

20. Predicables and Categories

Aristotle connects the four predicables with the ten categories.

1. Essence
2. Quantity
3. Quality
4. Relation
5. Place
6. Time
7. Position
8. State
9. Activity
10. Passivity

Accident, genus, property, and definition will always fall into one of these categories.

Key Distinction

Predicables = how a predicate belongs

Categories = what kind of predicate it is

Teaching Point

Categories classify kinds of being. Predicables classify modes of predication.

Categories answer “what kind?” Predicables answer “how does it belong?”

21. Dialectical Propositions

A dialectical proposition is not just any proposition.

It is a question asking something held by all, most, or the philosophers, provided that it is not contrary to general opinion.

Dialectical proposition = reputable claim suitable for argument

Examples

Is the knowledge of contraries the same?

Should one do good to one’s friends?

Dialectical propositions also include views like generally accepted opinions, contradictories of contraries of generally accepted opinions, and claims in accordance with recognized arts.

Teaching Point

Dialectic begins from claims an interlocutor can reasonably be expected to grant.

A dialectical premise must be reputable enough to be usable.

22. Dialectical Problems

A dialectical problem is a subject of inquiry that contributes either to:

choice and avoidance

or to:

truth and knowledge

It may contribute directly, or as a help to some other problem.

Examples

For choice and avoidance:

Is pleasure to be chosen or not?

For truth and knowledge:

Is the universe eternal or not?

Teaching Point

A dialectical problem must be genuinely disputable and worth inquiry.

Dialectic studies questions where reasoned disagreement is possible.

23. Theses

A thesis is a supposition of an eminent philosopher that conflicts with general opinion.

Thesis = reputable philosophical claim contrary to common opinion

Examples

Heraclitus: all things are in motion.

Melissus: Being is one.

A thesis is a kind of problem, but not every problem is a thesis.

Thesis $\subset$ Problem

Teaching Point

A thesis is worth examining because it is both serious and controversial.

Dialectic gives disciplined form to philosophical controversy.

24. What Should Not Be Examined Dialectically

Aristotle says not every problem or thesis should be examined.

Some questions are morally corrupt rather than dialectically puzzling.

Should one honor the gods and love one's parents?

Such a person needs correction, not argument.

Other questions require perception rather than dialectic.

Is snow white?

Such a person needs to look, not debate.

Teaching Point

Dialectic is for genuine puzzles, not for moral blindness or perceptual ignorance.

Not every question deserves dialectical treatment.

25. Dialectical Reasoning and Induction

Aristotle distinguishes two species of dialectical argument.

1. Reasoning
2. Induction

Reasoning

Reasoning proceeds from premises to a necessary conclusion.

premises $\rightarrow$ necessary conclusion

Induction

Induction passes from individuals to universals.

particulars $\rightarrow$ universal

Example

The skilled pilot is best at his task.

The skilled charioteer is best at his task.

$\therefore$ In general, the skilled person is best at his task.

Teaching Point

Induction is clearer and more persuasive to most people, while reasoning is more forceful against contradictory people.

Dialectic uses both examples that persuade and arguments that compel.

26. The Four Sources of Arguments

Aristotle explains how we become well supplied with arguments.

There are four sources.

1. Securing propositions
2. Distinguishing the senses of terms
3. Discovering differences
4. Investigating likenesses

Teaching Point

To argue well, one must prepare materials before the argument begins.

Dialectic requires a supply of propositions, meanings, differences, and likenesses.

27. Securing Propositions

Aristotle recommends collecting propositions from several sources.

1. opinions held by all
2. opinions held by most
3. opinions held by philosophers
4. opinions contrary to what seems generally accepted
5. opinions in accordance with recognized arts
6. opinions like those generally accepted

Practical Method

Draw up lists under headings.

On Good

On Life

On Pleasure

On Justice

Also note the views of individual thinkers.

Empedocles says the elements of bodies are four.

Teaching Point

Dialectic requires preparation: collect and organize argumentative material.

The dialectician is well supplied because he has already gathered endoxa.

28. Ethical, Natural, and Logical Propositions

Aristotle roughly divides propositions and problems into three kinds.

1. Ethical
2. Natural-philosophical
3. Logical

Ethical Example

Should one obey one's parents or the laws, if they disagree?

Natural-Philosophical Example

Is the universe eternal or not?

Logical Example

Is the knowledge of opposites the same or not?

Teaching Point

Dialectic can discuss many subjects because it works from reputable opinion rather than from the proper principles of a single science.

Dialectic ranges widely because its starting points are common opinions.

29. Philosophy and Dialectic Treat Propositions Differently

For philosophy, propositions should be treated according to truth.

For dialectic, propositions are treated with an eye to general opinion.

Philosophy $\rightarrow$ truth

Dialectic $\rightarrow$ general opinion

Teaching Point

Dialectic is not indifferent to truth, but its immediate method is to reason from what is accepted.

Dialectic is truth-seeking through reputable opinion.

30. Distinguishing Ambiguous Meanings

Aristotle says we must distinguish how many senses a term bears.

It is not enough to list the senses. We should also try to define them.

Example: Good

The term “good” may be said in different senses.

virtue is good

what produces health is good

what produces vigor is good

In one case, good is said because of an intrinsic quality. In another, it is said because something produces a result.

Teaching Point

One word may hide many questions.

Clarifying meanings prevents arguments from being aimed merely at words.

31. Tests for Ambiguity

Aristotle gives many ways to test whether a term has more than one meaning.

1. Examine its contraries
2. Examine intermediates
3. Examine contradictory opposites
4. Examine privation and possession
5. Examine inflected forms
6. Examine categories
7. Examine genera
8. Examine definitions in combination
9. Ask whether it admits of more and less
10. Examine differentiae

Teaching Point

Ambiguity is discovered by testing the relations around a term.

A word's meaning is revealed by its logical neighborhood.

32. Ambiguity Through Contraries

A term may be ambiguous if its contrary differs in different contexts.

Example

“Sharp” has different contraries.

sharp note $\leftrightarrow$ flat note

sharp edge $\leftrightarrow$ dull edge

Therefore “sharp” is not being used in the same sense in both cases.

Teaching Point

Different contraries often reveal different meanings.

If the opposites differ, the original term may be ambiguous.

33. Ambiguity Through Categories

A term may be ambiguous if it signifies different categories in different uses.

Example

“Good” may signify:

quality, as in a virtuous soul

relation to health, as in good medicine

time, as in the right occasion

quantity, as in the proper amount

Teaching Point

If the same word ranges across categories, it may not have one meaning.

Category shifts often signal ambiguity.

34. Ambiguity Through Genera

A term may be ambiguous if the things named by it belong to different genera that are not subaltern.

Example

“Donkey” may refer to:

an animal

or:

an engine

Since these belong to different genera, the term is ambiguous.

Teaching Point

Different non-subaltern genera reveal different definitions.

If the genera differ, the meaning probably differs.

35. Why Ambiguity Matters

Distinguishing meanings is useful for several reasons.

1. It promotes clarity.
2. It prevents reasoning merely about words.
3. It prevents being misled by false reasoning.
4. It helps expose an opponent’s equivocation.

Teaching Point

Dialectic requires clarity about what is being discussed.

If the meaning is unclear, the argument may not have a single subject.

36. Differences

Aristotle says we should examine the differences between things.

Differences should especially be examined within the same genus.

Examples Within the Same Genus

How does justice differ from courage?

How does wisdom differ from temperance?

These all belong to the genus of virtue.

Example Across Genera

We may also compare across genera if they are not too far apart.

How does sensation differ from knowledge?

Teaching Point

To know what a thing is, know how it differs from its neighbors.

Difference helps with sameness, definition, and classification.

37. Likenesses

Aristotle also says we should investigate likenesses.

Likeness is especially useful among things belonging to different genera.

Analogical Form

$$A : B = C : D$$

Example

knowledge : object of knowledge = sensation : object of sensation

Another example:

sight is in the eye as reason is in the soul

Teaching Point

Likeness lets us transfer insight from one case to another.

Analogy supplies arguments across different domains.

38. Uses of Likeness

The examination of likeness is useful for:

1. inductive arguments
2. hypothetical reasonings
3. rendering definitions

Induction

Induction depends on recognizing likeness among particular cases.

similar cases $\rightarrow$ universal claim

Hypothetical Reasoning

If it is generally accepted that what is true of one similar case is true of the others, then we may use likeness to reason by hypothesis.

Definition

Likeness helps us see the common genus or common feature.

point : line = unit : number

Each is a kind of starting point.

Teaching Point

Likeness is not merely decorative. It is a tool for finding universals, hypotheses, and definitions.

Similarity helps the dialectician move from case to principle.

39. The Architecture of Book I

Book I introduces the whole project of the *Topics*.

program of dialectic $\rightarrow$ kinds of reasoning $\rightarrow$ uses of dialectic $\rightarrow$ propositions and problems

$\rightarrow$ four predicables $\rightarrow$ sameness $\rightarrow$ categories $\rightarrow$ dialectical propositions and problems

$\rightarrow$ reasoning and induction $\rightarrow$ four sources of arguments

Teaching Point

Book I is not yet the full collection of topical rules. It is the introduction that explains what dialectic is, what it is for, what its materials are, and how arguments are supplied.

Book I gives the map of dialectical inquiry.

40. Summary of Main Ideas

1. The *Topics* studies dialectical reasoning from generally accepted opinions.
2. Reasoning is an argument in which something follows necessarily from what has been laid down.
3. Demonstrative reasoning begins from true and primary premises.
4. Dialectical reasoning begins from reputable opinions.
5. Contentious reasoning begins from merely apparent reputable opinions or merely appears to reason.
6. Dialectic is useful for intellectual training, casual encounters, and philosophical inquiry.
7. Dialectic is especially important for examining first principles.
8. Propositions provide the materials of argument; problems provide the subjects of inquiry.
9. Every dialectical problem concerns definition, property, genus, or accident.
10. Definition signifies essence.
11. Property is convertible with the subject but does not signify essence.
12. Genus is predicated in the category of essence of things differing in kind.
13. Accident may belong or not belong to the same thing.
14. The predicables classify how predicates belong; the categories classify what kind of predicate is involved.
15. Dialectical propositions and problems must be reputable, disputable, and useful.
16. Induction moves from particulars to universals; reasoning moves from premises to necessary conclusion.
17. The four sources of arguments are propositions, meanings, differences, and likenesses.
18. Ambiguity must be carefully detected and resolved.
19. Differences help define and distinguish things.
20. Likenesses support induction, hypothesis, and definition.

Final Framing

Topics, Book I, teaches the art of finding arguments from reputable opinions.

The *Posterior Analytics* showed how science demonstrates from first principles. The *Topics* shows how philosophy trains itself toward principles by testing, organizing, and arguing from what is generally accepted.

Dialectic is disciplined reasoning in the space between opinion and science.